



OpenRiverCam in Indonesia: What the Pilot Taught Us, and the Path Forward

Indonesia ORC Deployment — Spring 2026

PMI · IPB · BHLK · American Red Cross

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Partners: Palang Merah Indonesia · Institut Pertanian Bogor · Balai Hidrologi dan Lingkungan Keairan · American Red Cross

Prepared for: IPB and BHLK, following the PMI / IPB / BHLK meeting at Sukabumi, 21 August 2026.

Reading time: about eight minutes, plus the index at the back.

Companion documents: the Appendix, which holds the measurements, interface specifications and procedures behind everything below; and `docs/RECOMMENDATIONS.md`, the working list this report is drawn from.

Thank you

None of this exists without the people who built it and looked after it. PMI volunteers and staff at Sukabumi and Jakarta gave their time to a system that was new to them and that did not always work. IPB re-surveyed the site with a total station after our own survey failed twice, and that survey is what the station runs on today. BHLK brought the standards knowledge, and the offer of server capacity, that make a pilot possible at all.

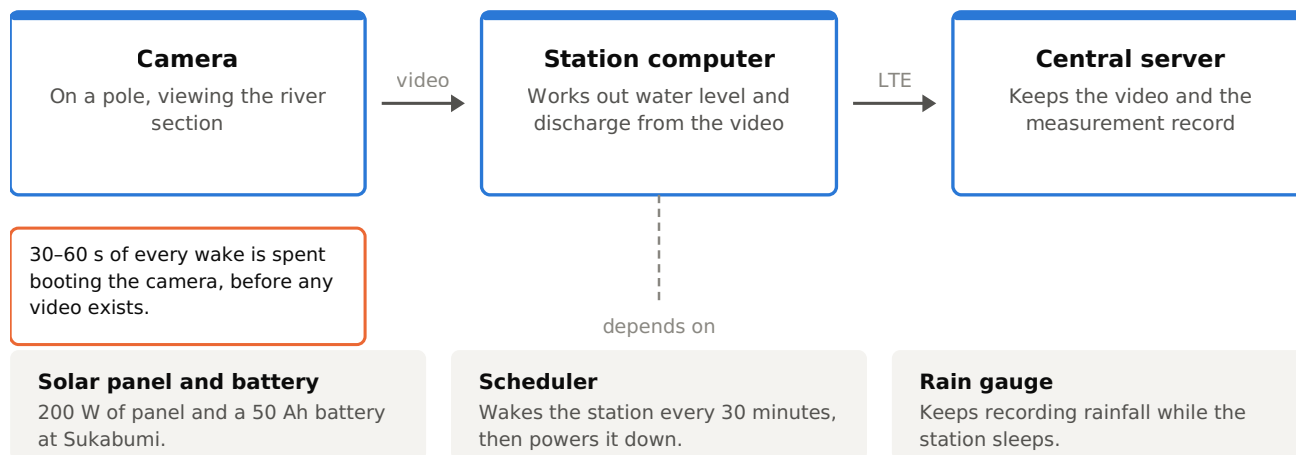
What follows is largely a list of things we would do differently. Every one of them was learned because someone did the work that made it visible.

What we did

A camera on a pole watches the river. A small computer in a weatherproof box turns the video into a water level and a discharge figure, and a mobile data link sends both, with the video, to a server. Sukabumi runs on solar power and wakes every thirty minutes: it starts, records, processes, uploads and shuts down.

Figure 1 How a measurement is made

The path a single five-second video takes, from the river to the record.



The station is awake about two minutes in every thirty. Everything above happens inside that window.

The camera on its pole sends video to the station computer, which derives water level and discharge; both travel over the mobile network to the server. The three things the computer depends on are shown beneath it. The whole sequence happens inside a waking period of about two minutes, of which the first thirty to sixty seconds is spent waiting for the camera to start.

The station came to **USD 1,340 in materials** — electronics and enclosure only, since Sukabumi already had a solar panel and battery on site. For comparison, the lowest-cost automatic water-level station we could confirm on the Indonesian government e-catalogue is about **USD 3,600 before VAT**, and it measures stage only. That price difference is the whole argument for studying this design: how many stations an agency can afford decides how much of a catchment it can observe.

Five constraints governed every choice of part, and they are why the design can be copied at all: commodity parts with more than one supplier; no soldering, every connection a screw terminal, plug or header; no specialist assembly skills; common hand tools; and any part replaceable in five minutes.

Two stations were built. Sukabumi was installed in April and has run since. The Jakarta station was finished and tested, but permission for its intended site did not arrive, and it is still at Wisma PMI, unpowered since April.

What we learned

Sukabumi is a volunteer-supported pilot. It is visited rarely and looked after remotely by people doing it alongside other work. It was never built to production standards of availability, and measuring it against an industrial instrument would be the wrong test. The useful question is narrower: what does this design make hard?

A twenty-five dollar decision caused most of the outages, and no single step in the chain looks

like a mistake. The USB storage drive caused a driver fault at boot, so it was removed rather than fixed. That left the SD card as the only volume, small enough to sit permanently at the level where it deletes old recordings. That caused processing to fail on 43% of videos. Because the station shuts down after processing finishes, a failed run meant it never shut down: it stayed awake to the scheduling board's 25-minute backstop instead of stopping after two minutes, roughly twelve times the energy for that cycle. Repeated across a night, that flattened the battery — and a missed wake left the next-startup alarm in the past with nothing to reset it, so one missed cycle became days.

Two lessons follow. **Shutdown and startup are controlled by two different systems and neither owns the whole cycle**, so when processing fails they fail together. And **cheap parts can carry expensive operating costs** — the camera is the other example, bought for about USD 60 against an alternative near USD 1,268, meeting its specification but running a reseller's firmware with capability removed, so recorded video cannot be fetched and capture falls back to a lower quality live stream.

There is a common root worth naming. The computer's own clock battery uses a small connector that **broke on both boards**. Losing it is why a separate scheduling board was fitted, which is what split the sleep and wake cycle across two systems in the first place. A computer with storage, power and timekeeping already integrated and protected would have removed that whole sequence, and it is the strongest recommendation we have.

The station cannot say what is wrong with it. No interruption announced itself. The record had to be reconstructed afterwards from the server database, and a fault the system never reports cannot be found by watching for it. Related: nothing compares what the



The parts for one station. Nothing is fabricated and nothing is single-source.

station recorded against what the server received, so data can go missing with no symptom at either end.

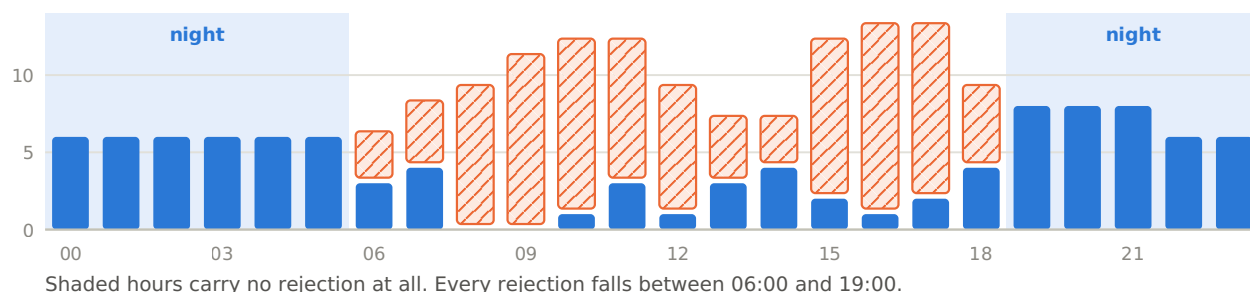
Water level is the weakest measurement, and it fails in daylight. The station reads the level from the video. In a 200-capture sample, every rejection fell between 06:00 and 19:00 and none at night. It matters more than a missing level because level is computed first — when it fails the whole measurement is discarded, including the surface velocity that succeeded — and because level feeds the cross-sectional area, so an error in it scales the discharge figure directly.

Figure 2 Reading the water level from video, 8 to 14 July 2026

200 captures. A rejection costs the whole discharge measurement, not only the water level.

■ Water level accepted (100) ▨ Rejected — not confident enough (100)

A By hour of day (local time, WIB)



B How confident the reading was, for all 200 captures



Rejected captures are not near misses: half sit at 1.63 or below, and only 23 of the 100 reach 1.8.

Accepted captures spread well above the threshold, though 36 of them sit between 2.0 and 3.0.

Captures counted by hour of day, and how confident each reading was against the threshold at which a water level is accepted. Rejections peak in mid-morning and again in mid-afternoon, with fewer in the early afternoon — the pattern a sun-angle effect produces.

The survey is unforgiving, and IPB fixed it. Two RTK surveys on consecutive days, with the same equipment and crew, disagreed with themselves by about 99 cm horizontally and 139 cm vertically. Repeating a method that has failed reproduces the same noise. IPB's total-station survey replaced the approach and is what the station runs on today, at 3.7 cm RMSE.

We had no station we could break. Everything above was diagnosed on a solar station, on a river, awake for tens of seconds at a time, that we could not touch. We could not

reproduce a fault, test a fix before committing it to a remote machine, or tell the difference between a change working and the fault simply not happening that week. Several of these diagnoses took months for that reason alone. The Jakarta station was meant to be that test station and never became one.

What we recommend

The full set is indexed at the back and detailed in the working list. The headlines:

Use an industrial Raspberry Pi carrier with storage, power and timekeeping already integrated. We would advocate for this strongly, because most of the failures above meet at one place. The Pi's own clock battery uses a small connector that broke on both boards; losing it is why a separate scheduling board was added, which is what split shutdown and startup across two systems. And the USB drive fault left the operating system and all video on the SD card, which is what filled the disk. A carrier with NVMe on board, a protected battery-backed clock, wide-input power with an integrated UPS that reports voltage and current, and often a modem slot, removes all of that in a single part. It costs more per unit and it is a single-source component, which is a real trade against the commodity-parts rule — but it is done by people who do it repeatedly, and it is the change that would have prevented the largest number of our problems. **R37**

Build the test station first. A working station, mains-powered, always on, somewhere someone can watch it, open it and break it deliberately. It is where a fault is reproduced before it is diagnosed remotely, where a software change is tried before it goes to a river, and where a server upgrade is rehearsed before it obliges every station to follow. We would treat it as the first station a pilot builds, not the last. **R36**

Build monitoring for a fleet, not for one station. What ORC-OS reports is enough for a single station watched by the people who built it. An agency running ten or fifty will need more: whether each station woke and shut down, time awake per cycle, voltage and current together, free disk space against its deletion threshold, captured against delivered, processing outcome per video, and last contact per station. This is the highest-value work on the list. **R4-R7**

Fit an independent water-level reference, and plan the survey as skilled work from the start. A level sensor or a staff gauge in the camera's view removes the dependence on one optical method. And carry an independent field check of a different kind, so a survey problem is found before the team leaves site. **R1, R2, R19, R20**

Give one process control of the whole sleep and wake cycle, set the next wake as part of shutting down, and make shutdown happen on a timer regardless of what processing does. **R10, R12**

Make mains power the default, and use solar only where mains is not available.

An always-on station removes the wake cycle and, with it, most of what is described above.

R11**Choose the site before anything else, and confirm permission in writing before building for it.**

The site fixes limits nothing later can undo: satellite positioning for the survey, sun angle for the water level, and how much of the flow the camera resolves.

R16-R18

Keep what works. The five constraints above all. And **keep the spare switched outputs** — the relay module has four channels, one drives the camera and three are deliberately left free, wired and ready. That is what lets a station drive a siren, a beacon or an alerting relay without reopening the design, and it is what turns a measurement station into something a community can act on. We would ask that any replication keeps that spare capacity rather than removing it to save a few dollars. **R30-R35**

The path forward

Our recommendation is that the next step is yours. The pilot produced knowledge rather than a body of measurements, and knowledge is worth more as an input to your own design than as a set of parts to copy. We would rather help you start your own approach than hand you ours to maintain.

The Jakarta station is available and we would suggest a particular use for it: a study and test unit. Open it, trace it, power it up, take it apart, and install it locally if that is useful. What we would not recommend is putting it into service as an operational station with expectations of availability, because it carries the design this report recommends changing.

We are not proposing to build your stations. What we can offer is the record — this report, the appendix, the operator and assembly documentation in English and Bahasa Indonesia, the software, and the built station to take apart — along with whatever else is useful: reviewing a design, answering a question about something that surprised us, looking at data that does not behave. Sukabumi will keep running and we will keep reporting what it does, including the parts that go wrong.

Questions for consideration

- Who should hold the Jakarta station as a study and test unit, IPB or BHLK?
- Can a staff gauge be read from the camera image accurately enough? If it can, R1 needs no separate sensor.
- Two interruptions remain unexplained, with the energy signature of the maintenance setting without that setting being on.

- How does the velocity measurement perform across the surface conditions the pilot sites present? This has not been characterised.
- Absolute discharge accuracy at Sukabumi, unresolved pending further survey work.

We expect your work to change some of what is written here.

Index of recommendations

The full text of each is in `docs/RECOMMENDATIONS.md`. Measurements and procedures are in the companion appendix.

What is worth keeping

	Recommendation
R30	Keep the five constraints
R31	Keep spare switched outputs, and design for community alerting
R32	Keep the camera factory-sealed
R33	Keep the recovery kit with the station
R34	Keep aligning with existing practice
R35	Keep the software stack open

Keep a station you can break

	Recommendation
R36	Put a test station on a bench, online, and keep it there

Choose the site before anything else

	Recommendation
R16	Treat site selection as a measurement decision, not a logistics one
R17	Confirm site permission in writing before any hardware is built for that site
R18	Engage local academic and basin partners in site selection

Improve measurement accuracy

	Recommendation
R1	Fit an independent water-level reference

	Recommendation
R2	Plan the survey as skilled work from the start, and budget for it
R19	If a survey method fails once at a site, change the method
R20	Carry an independent check of a different kind, and use it before leaving site

Improve monitoring

	Recommendation
R4	Build a monitoring regime for a fleet, not a station
R5	Have the station send diagnostics; do not require a login
R6	Record voltage and current together
R21	Remove the rain gauge from the camera station
R7	Compare what was recorded against what arrived, automatically

Give one process control of the sleep and wake cycle

	Recommendation
R10	One process should control both shutdown and startup
R12	A processing failure must not prevent shutdown

Improve the server and the processing chain

	Recommendation
R13	Manage disk space before it stops the station working
R14	A failed water level should not discard the whole measurement
R15	Monitor the server as well as the station
R22	Plan video storage before the fleet grows
R23	Configuration edits must go through one interface
R24	Do not move bulk media through the API

Choose parts for operating cost, not just purchase price

	Recommendation
R37	Use an industrial Raspberry Pi carrier with storage, power and timekeeping already integrated

	Recommendation
R9	These stations were built with low unit cost as the primary goal, and some of those choices raise operating cost at volume
R25	Three specific parts cost more than they saved
R26	Check the control interface before buying anything under software control

Choose the architecture to suit the site

	Recommendation
R11	Mains power should be the default. Use solar only where mains is not an option
R8	Consider putting the computer indoors, and decide deliberately where the processing happens
R27	A camera-only field node is the most deployable version of this system

Support local construction and maintenance

	Recommendation
R3	Specify interfaces, not part numbers
R28	Order locally-sourced parts weeks in advance, not on arrival

Divide responsibilities

	Recommendation
R29	Write the division of responsibility into the agreement

Supporting documentation

Appendix sections: camera firmware limitations and the capture path (A1); firmware replacement risks (A2); survey scope of work and acceptance checks (A3); the availability record (A4); the water-level dataset (A5); data delivery measurements (A6); and the power, scheduling and always-on comparisons (A7).

Operator and assembly documentation, in English and Bahasa Indonesia, is available on request.